SURVEYING & LEVELLING · NOTE SL12

Levelling Errors, Instrument Adjustments and Error of Closure

Course topics: SWCE1.4 L11

Care and precautions

- Carry the level in its case; never let it fall or leave it unguarded; protect tripods and rods from the weather.
- Handle the instrument carefully from the start — rough handling marks an unskilled surveyor.
- Lift it out of the case gently, with your fingers beneath the horizontal bar.

Adjustments of the level

Prolonged or rough use throws a level out of adjustment. Adjust it carefully, preferably under supervision. The parts adjusted are the cross hairs and the bubble tube.

Condition of perfect adjustment	Test and correction
The axis of rotation is truly vertical	Set up and level; place the telescope over one pair of opposite screws and centre the bubble; rotate 180°. If the bubble moves, bring it halfway back with the bubble-tube screws
The line of sight is horizontal (parallel to the level-tube axis) when the bubble is central	**Two-peg test:** drive pegs A and B about 60 m apart. Set up midway and read a on A and b on B — the true difference in elevation is b - a. Set up near A and read a′ and b′. If b′ - a′ = b - a the level is in adjustment; otherwise move the cross-hair ring until it is
The horizontal cross hair lies in a horizontal plane	Sight a well-defined point and turn the telescope about the vertical axis. If the point leaves the middle hair, loosen the ring screws, rotate the ring slightly and retighten

Error of closure

After closing a level circuit, the difference between the computed final elevation and the known starting elevation is the **error of closure**. Run every bench level circuit back to its starting point so that this error can be found.

$$\text{Permissible error of closure} = \pm\, C\,\sqrt{K}\ (K = \text{length of the circuit in km})$$

Class of levelling	Common textbook limit
Rough levelling	± 0.100 √K m
Ordinary levelling	± 0.025 √K m
Accurate levelling	± 0.012 √K m
Precise levelling	± 0.006 √K m

Correction. The source notes give the limit for ordinary levelling as "±√(distance in kilometres) mm". The coefficient has dropped out: that would allow only 1 mm over a 1 km circuit — stricter than precise levelling. Standard Indian surveying texts (e.g. B. C. Punmia, Surveying Vol. 1) give the class limits in the table above; the precise-levelling value of ±0.006√K m also appears in exam keys.

Sources of error in levelling

Error	How to keep it small
Bubble not central at the moment of reading	Check the bubble before every reading
Parallax	Focus the eye-piece on the hairs, then the objective on the staff
Staff not vertical	Use the vertical hair; signal the rodman to plumb the rod
Line of sight not horizontal (collimation error)	Keep BS and FS distances about equal; adjust by the two-peg test
Turning point sinks or moves	Choose firm turning points
Reading or booking mistakes	Apply the arithmetic checks; close every circuit

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. In the two-peg test, the pegs are set about

(a) 15 m apart
(b) 30 m apart
(c) 60 m apart
(d) 200 m apart

Q2. If, after rotating the telescope 180°, the bubble moves off centre, correct it by

(a) the full amount with the foot screws
(b) half with the bubble-tube screws
(c) moving the diaphragm
(d) replacing the tube

Q3. Permissible closing error in levelling varies as

(a) K
(b) √K
(c) K^2
(d) 1/K

Q4. Keeping back-sight and fore-sight distances equal mainly eliminates

(a) parallax
(b) collimation error
(c) staff tilt
(d) booking errors

Answers

Q1 — (c) 60 m apart
Q2 — (b) half with the bubble-tube screws
Q3 — (b) √K — *K = circuit length in km.*
Q4 — (b) collimation error